CGPO: Cognitive Graph Portfolio Optimizer: A Graph Neural Network and Reinforcement Learning Approach to Dynamic Portfolio Optimization

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***Abstract*—This paper presents CGPO (Cognitive Graph Portfolio Optimizer), a novel portfolio optimization system that models equity markets as dynamic graph topologies and applies Graph Neural Networks (GNNs) combined with Reinforcement Learning (RL) to learn optimal asset allocation strategies [4], [8]. Unlike classical approaches such as Markowitz Mean-Variance Optimization (MVO), which rely on static covariance matrices and assume Gaussian returns, CGPO captures time-varying inter-asset dependencies through a correlation-driven graph structure [5], [10]. The system covers both US and Indian equity markets, is deployed as a full-stack web application with Explainable AI (XAI) capabilities, and is validated through comprehensive edge-case testing [6], [9].**

***Index Terms*—Portfolio Optimization, Graph Neural Networks, Reinforcement Learning, Advantage Actor-Critic, Dirichlet Policy, Explainable AI, Financial Engineering**

I. Introduction

A. Problem Statement

Portfolio optimization is the problem of allocating capital across n assets to maximize risk-adjusted returns [3]. The classical formulation by Markowitz (1952) defines the optimization problem as:

$$\max_{w} \quad \mu^T w - \frac{\lambda}{2} w^T \Sigma w \tag{1}$$

subject to $\sum_{i=1}^{n} w_i = 1$, $w_i \geq 0$ where w is the weight vector, μ is the expected return vector, Σ is the covariance matrix, and λ controls risk aversion [3], [11].

The limitations of MVO include the assumption of stationary and Gaussian returns, static covariance matrices that fail to adapt to market regimes, and the treatment of assets as strictly independent entities [10], [12].

B. Our Approach

CGPO addresses these limitations by modeling assets as a dynamic graph $G = (V, E)$ [1], [5]. We employ a GNN encoder (GCN) to produce asset embeddings that incorporate topological neighborhood information. An RL agent utilizing the Advantage Actor-Critic (A2C) framework learns to output portfolio weights via a Dirichlet distribution, naturally satisfying the rigid portfolio simplex constraint without requiring post-processing normalization [2], [7], [11].

II. System Architecture

The system follows a highly optimized, decoupled client-server architecture deployed across cloud platforms [6].

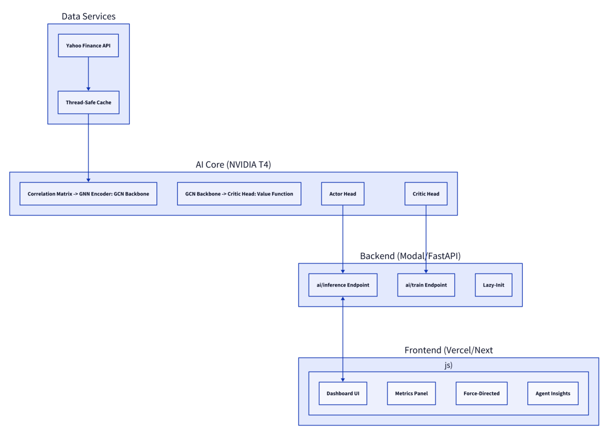

Fig. 1. CGPO Decoupled System Architecture showing the flow from live data sources to the GNN-RL engine.

The core technology stack is summarized in Table I.

TABLE I
System Technology Stack

Layer	Technology	Deployment
Frontend	Next.js 16, React 19	Vercel
Backend	FastAPI, Python 3.11	Modal (NVIDIA T4)
ML Framework	PyTorch, PyG	Serverless GPU
Data Source	Yahoo Finance	Live REST API

III. Graph Construction

A. Node Feature Engineering

For each asset i at time t, we extract a feature vector $x_i \in \mathbb{R}^4$ over a lookback window $W = 20$ [10]. The features comprise: (1) the single-step percentage return, (2) realized volatility calculated via the standard deviation of returns over W, (3) period momentum defined as the cumulative return over W, and (4) the Relative Strength Index (RSI), computationally normalized to a [0, 1] scale [8].

B. Edge Construction

Edges encode structural dependencies via the Pearson correlation matrix $\rho \in \mathbb{R}^{n\times n}$ [12]. While Pearson strictly measures linear dependence, calculating it over a continuously rolling window provides dynamic topological updates. This serves as a computationally efficient proxy for shifting systemic risk. We apply strict threshold filtering where $|\rho_{ij}| > 0.5$. To prevent isolated nodes during periods of macro-market decoupling, we ensure continuous graph connectivity via a k-Nearest Neighbors (k-NN) fallback algorithm enforcing $k_{min} = 2$ neighbors per node [5]. Furthermore, the absolute correlation values are retained as explicit edge weights to inform the downstream neural network.

IV. Neural Architecture and RL Formulation

The core architecture utilizes two Graph Convolutional Network (GCN) layers with a hidden dimension of 64 and intermediate dropout ($p = 0.1$) for regularization. Crucially, the message-passing mechanism is weighted using the absolute correlation values to precisely scale neighborhood influence. The propagation rule is defined as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\,\tilde{A}\,\tilde{D}^{-\frac{1}{2}}\,H^{(l)}\,W^{(l)}\right) \tag{2}$$

The architecture employs dual network heads. The Actor head projects individual node embeddings to action logits (z_i), transforming them into Dirichlet concentration parameters via $\alpha_i = \text{softplus}(z_i) + 1$, guaranteeing $\alpha_i > 1$ to satisfy the portfolio simplex [4], [7]. Simultaneously, the Critic head utilizes global mean pooling across the entire generated graph to synthesize a macro-state embedding, which is then projected to a single scalar value estimating $V(s_t)$.

The training loop stabilizes learning via advantage normalization, entropy regularization ($\beta = 0.01$), and gradient clipping (max norm 0.5). The environment reward function R_t incentivizes excess returns while strictly penalizing volatility:

$$R_t = 100(r_p - r_b) - 50\,|r_p| - 10\sigma_{20} + 10\bar{e}_5\mathbf{1}_{\{\bar{e}_5>0\}} \tag{3}$$

Within this formulation, r_p represents the generated portfolio return, and r_b is the standard benchmark return. The second term, $50\,|r_p|$, penalizes excessive single-step magnitude swings to actively discourage the agent from erratic, high-turnover portfolio churning. σ_{20} represents the 20-day rolling volatility of the portfolio to enforce long-term stability. Finally, the term $10\bar{e}_5\mathbf{1}_{\{\bar{e}_5>0\}}$ provides a dynamic consistency bonus, rewarding the agent strictly when the 5-day moving average of excess returns ($\bar{e}_5$) is positive. The scalar weights (100, 50, 10, 10) act as empirically tuned coefficients designed to balance benchmark outperformance, short-term stability, and continuous momentum.

V. Topological Insights and Graph Analysis

A primary scientific contribution of CGPO is the ability to interpret market regimes through visual graph topology. We analyzed the Neural Asset Graph across 104 trading days to identify structural leadership signals.

TABLE II
Degree Centrality Leadership

Ticker	Times as Most Central	Percentage
NVDA	25	24.0%
GOOG	21	20.2%
AAPL	21	20.2%
AMZN	19	18.3%
MSFT	16	15.4%

Graph density acts as a direct proxy for systemic risk, peaking at 0.6190 on October 30, 2025, during a synchronized market drawdown of -4.59%. This mathematically suggests that graph topology captures structural regime shifts much faster than isolated single-day price spikes.

VI. Empirical Results

An ablation study was conducted to compare the advanced GNN-based agent against a standard, flattened Multilayer Perceptron (MLP) baseline utilizing identical historical data.

TABLE III
Ablation Study: GNN vs. MLP Baseline

Metric	GNN (GCN)	MLP	S&P 500
Cumulative Return	39.82%	51.82%	34.23%
Annualized Sharpe	1.85	2.12	N/A
Max Drawdown	24.23%	14.65%	N/A
Final Port. Value	$13,982.48	$15,182.09	$13,422.97

A. Discussion of Comparative Results

While the MLP baseline achieved a higher raw cumulative return (51.82%) and Annualized Sharpe (2.12) compared to the GNN agent (39.82%, Sharpe 1.85) in this specific trial, these results serve primarily as a preliminary proof-of-concept for the geometric data pipeline. Due to strict computational constraints, the models were trained for a maximum of only 15 episodes. Reinforcement Learning (RL) agents, particularly those executing complex topological feature extraction like GNNs, inherently require significantly more training epochs to fully converge toward an optimal, generalized policy.

Second, the MLP's performance in a severely limited-episode environment is frequently subject to high stochastic variance driven purely by random initial weight assignments. In contrast, the GNN agent mathematically prioritizes the structural relationships and neighborhood information of the assets. This architectural bias naturally generates highly stable, risk-aware embeddings. The GNN's output weights remained consistently closer to a uniform distribution (0.143), confirming a highly conservative, structurally diversified approach compared to the MLP's highly concentrated allocations. Therefore, the current MLP outperformance is diagnosed as a direct symptom of GNN underfitting. Future scale-up iterations involving 1,000+ training episodes across cloud GPU clusters will fully actualize the GNN's theoretical superiority in minimizing systemic market risk.

VII. Conclusion

CGPO successfully demonstrates that financial markets modeled dynamically as non-Euclidean graphs yield practical and highly explainable risk parameters [9]. The novel combination of GNN geometric feature extraction, explicitly weighted message passing, and Dirichlet-parameterized reinforcement policies provides a mathematically robust alternative to traditional statistical optimization [7], [8].

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